

Status	Beendet
Begonnen	Samstag, 24. Januar 2026, 14:15
Abgeschlossen	Samstag, 24. Januar 2026, 20:50
Dauer	6 Stunden 35 Minuten
Punkte	47,00/49,00
Bewertung	9,59 von 10,00 (95,92%)

Frage **1**

Richtig

Erreichte Punkte 1,00 von 1,00

Query compilation requires fewer CPU instructions and hides cache misses better than vectorized query processing.

☒ Wahr ☐ Falsch

Die richtige Antwort ist 'Wahr'.

Frage 2

Richtig

Erreichte Punkte 1,00 von 1,00

Which of the following processing models offers a balanced combination of data and instruction caching?

Wählen Sie eine Antwort:

- ☒ vector-at-a-time ✓
- ☐ tuple-at-a-time
- ☐ operator-at-a-time

Your answer is correct.

Die richtige Antwort ist: vector-at-a-time

Frage 3

Richtig

Erreichte Punkte 1,00 von 1,00

The five base operators of the relational algebra are selection, projection, intersect, union, and cartesian product.

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage 4

Richtig

Erreichte Punkte 1,00 von 1,00

The set difference of two relations can be expressed using only set intersection operators.

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage 5

Richtig

Erreichte Punkte 1,00 von 1,00

The grouping operator γ is used to filter tuples based on a condition.

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage 6

Richtig

Erreichte Punkte 1,00 von 1,00

A join is the composition of a set intersection and a selection operator.

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage 7

Falsch

Erreichte Punkte 0,00 von 1,00

Selection in relational algebra is used to retrieve only certain attributes from a relation.

- ☒ Wahr ✗
- ☐ Falsch

The selection operator filters rows; the projection operator selects which attributes exist in the output relation.

Die richtige Antwort ist 'Falsch'.

Frage 8

Richtig

Erreichte Punkte 1,00 von 1,00

Grouping, sort and join are all pipelining operators.

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage 9

Richtig

Erreichte Punkte 1,00 von 1,00

Given the following SQL query and execution steps, calculate the number of invocations per operator for each processing model (Volcano, Materialization, Vectorization).

SQL:

```
SELECT MIN(e.salary)
FROM employee e
WHERE e.age > 35;
```

Execution steps:

1. Scan table **employee e**.
2. Filter results on **e.age > 35**.
3. Aggregate tuples on **MIN(e.salary)**.

Important assumptions:

- Table employee consists of 500000 rows.
- The predicate (**e.age > 35**) has a selectivity factor of 0.5 (50% pass this filter).
- The vectorization model uses a batch size of 2000 tuples (processes 2000 tuples at a time).
- For the materialization model, intermediate results are fully materialized.

Volcano Model

Scan: ✓

Filter: ✓

Aggregate: ✓

Materialization Model

Scan: ✓

Filter: ✓

Aggregate: ✓

Vectorization Model

Scan: ✓

Filter: ✓

Aggregate: ✓

Frage **10**

Richtig

Erreichte Punkte 1,00 von 1,00

Is the following law valid?

$$\sigma_{C1 \text{ OR } C2}(R) = (\sigma_{C1}(R)) \cup (\sigma_{C2}(R))$$

Assume relation R is a set.

- ☒ Wahr 
- ☐ Falsch

Die richtige Antwort ist 'Wahr'.

Frage 11

Richtig

Erreichte Punkte 1,00 von 1,00

Consider four relations $R(b, c)$, $S(c, d)$, $T(a, d)$, and $U(a, b)$. Each relation has 1000 tuples. The relations have the following attribute domain cardinalities:

$R(b, c)$	$S(c, d)$	$T(a, d)$	$U(a, b)$
		$V(T, a) = 250$	$V(U, a) = 100$
$V(R, b) = 40$			$V(U, b) = 20$
$V(R, c) = 100$	$V(S, c) = 200$		
	$V(S, d) = 200$	$V(T, d) = 1000$	

For each join order: Can it be ignored because it contains a cross product when determining the optimal join order?

True	False			
☐	☒	$(R \bowtie S) \bowtie T$	☒	
☒	☐	$(R \bowtie T) \bowtie S$	☒	
☒	☐	$(S \bowtie U) \bowtie R$	☒	
☐	☒	$(R \bowtie S) \bowtie U$	☒	
☒	☐	$(R \bowtie T) \bowtie U$	☒	
☐	☒	$(R \bowtie U) \bowtie S$	☒	

 $(R \bowtie S) \bowtie T$

: False

 $(R \bowtie T) \bowtie S$

: True

 $(S \bowtie U) \bowtie R$

: True

 $(R \bowtie S) \bowtie U$

: False

 $(R \bowtie T) \bowtie U$

: True

$$(R \bowtie U) \bowtie S$$

: False

Frage 12

Richtig

Erreichte Punkte 1,00 von 1,00

Consider four relations $R(a, d)$, $S(b, c)$, $T(a, b)$, and $U(c, d)$. Each relation has 1000 tuples. The relations have the following attribute domain cardinalities:

$R(a, d)$	$S(b, c)$	$T(a, b)$	$U(c, d)$
$V(R, a) = 1000$		$V(T, a) = 10$	
	$V(S, b) = 200$	$V(T, b) = 200$	
	$V(S, c) = 40$		$V(U, c) = 200$
$V(R, d) = 100$			$V(U, d) = 250$

Select the best join order (the one with the lowest cost).

Note: Cost refers to the sum of all intermediate result sizes (excluding input relations and the final result).

Wählen Sie eine Antwort:

- ☐ $((R \bowtie U) \bowtie S) \bowtie T$
- ☒ $((T \bowtie R) \bowtie U) \bowtie S$ ✓
- ☐ $((S \bowtie U) \bowtie T) \bowtie R$
- ☐ $((U \bowtie T) \bowtie S) \bowtie R$

Your answer is correct.

Die richtige Antwort ist: $((T \bowtie R) \bowtie U) \bowtie S$

Frage 13

Richtig

Erreichte Punkte 1,00 von 1,00

Consider four relations $R(b, c)$, $S(a, b)$, $T(c, d)$, and $U(a, d)$. Each relation has 1000 tuples. The relations have the following attribute domain cardinalities:

$R(b, c)$	$S(a, b)$	$T(c, d)$	$U(a, d)$
	$V(S, a) = 40$		$V(U, a) = 10$
$V(R, b) = 1000$	$V(S, b) = 200$		
$V(R, c) = 400$		$V(T, c) = 40$	
		$V(T, d) = 10$	$V(U, d) = 250$

What is the cost of the following join order $((R \bowtie T) \bowtie U) \bowtie S$?

Note: Cost refers to the sum of all intermediate result sizes (excluding input relations and the final result).

$Cost(((R \bowtie T) \bowtie U) \bowtie S) =$ 

Die richtige Antwort ist: 12500

Frage 14

Richtig

Erreichte Punkte 1,00 von 1,00

Given are the tables $R(\underline{r_1}, s_1)$ and $S(\underline{s_1}, s_2)$. $R. s_1$ is a foreign key to $S. s_1$. The tables have the following tuple and attribute domain cardinalities:

- $T(R) = 800$
- $T(S) = 300$
- $V(S, s_2) = 10$

Assume that $1 / V(R, f)$ of the tuples satisfy the comparison when an attribute is equated to a constant (e.g., $\sigma_{f=15}$)

How many tuples are selected by the join $R \bowtie \sigma_{s_2=84}(S)$?

Antwort:



Die richtige Antwort ist: 80

Frage 15

Richtig

Erreichte Punkte 1,00 von 1,00

Consider four relations $R(a, d)$, $S(b, c)$, $T(a, c)$, and $U(b, d)$. Each relation has 1000 tuples. The relations have the following attribute domain cardinalities:

$R(a, d)$	$S(b, c)$	$T(a, c)$	$U(b, d)$
$V(R, a) = 20$		$V(T, a) = 500$	
	$V(S, b) = 250$		$V(U, b) = 10$
	$V(S, c) = 1000$	$V(T, c) = 100$	
$V(R, d) = 250$			$V(U, d) = 500$

What is the cardinality of the following join order $(S \bowtie R) \bowtie U$?

$T((S \bowtie R) \bowtie U) =$ 

Die richtige Antwort ist: 8000

Frage 16

Richtig

Erreichte Punkte 1,00 von 1,00

Consider the natural join $R(b, c, d) \bowtie S(b, c, e) \bowtie U(a, b, c)$, and the following important statistics:

R(b, c, d)	S(b, c, e)	U(a, b, c)
$T(R) = 12000$	$T(S) = 42000$	$T(U) = 18000$
		$V(U, a) = 100$
$V(R, b) = 300$	$V(S, b) = 150$	$V(U, b) = 40$
$V(R, c) = 50$	$V(S, c) = 100$	$V(U, c) = 150$
$V(R, d) = 100$		
		$V(S, e) = 70$

What is the estimated cardinality of the natural join (the join on all common attributes)?

Antwort: 

Die richtige Antwort ist: 13440

Frage 17

Richtig

Erreichte Punkte 1,00 von 1,00

Given is the table $R(a, b)$ with the following tuple and attribute domain cardinalities:

- $T(R) = 172800$
- $V(R, a) = 80$
- $V(R, b) = 40$

Assume that:

- $1 / (3 * V(R, f))$ of the tuples satisfy the comparison when an attribute is equated to a constant (e.g., $\sigma_{f=15}$)
- Inequality (e.g., $\sigma_{f < 90}$) tends to retrieve one-fourth of the tuples
- For not equals (e.g., $\sigma_{f \neq 90}$), a fraction of $4 / V(R, f)$ number of the tuples will fail to meet the condition
- Predicates are independent

Note, single assumptions may deviate from the ones discussed during the lecture.

How many tuples are selected by the selection $\sigma_{a \neq 75 \text{ AND } b=20}(R)$?

Antwort: 

Die richtige Antwort ist: 1368

Frage 18

Richtig

Erreichte Punkte 1,00 von 1,00

Given is the table $R(a, b)$ with the following tuple and attribute domain cardinalities:

- $T(R) = 28800$
- $V(R, a) = 60$
- $V(R, b) = 30$

Assume that:

- $1 / V(R, f)$ of the tuples satisfy the comparison when an attribute is equated to a constant (e.g., $\sigma_{f=15}$)
- Inequality (e.g., $\sigma_{f < 90}$) tends to retrieve one-third of the tuples
- For not equals (e.g., $\sigma_{f \neq 90}$), a fraction of $5 / V(R, f)$ number of the tuples will fail to meet the condition
- Predicates are independent

Note, single assumptions may deviate from the ones discussed during the lecture.

How many tuples are selected by the selection $\sigma_{a=71 \text{ OR } b=90}(R)$?

Antwort:



Die richtige Antwort ist: 1424

Frage 19

Richtig

Erreichte Punkte 1,00 von 1,00

Which of the following is a valid Condition Rule in the grammar for simple subset of SQL?

True	False			
☐	☒	<code><Condition> ::= <Attribute> FOUND IN <Pattern></code>		
☒	☐	<code><Condition> ::= <Attribute> = <Attribute></code>		
☐	☒	<code><Condition> ::= <Attribute> AVAILABLE <Pattern></code>		
☒	☐	<code><Condition> ::= <Condition> AND <Condition></code>		

`<Condition> ::= <Attribute> FOUND IN <Pattern>`: False
`<Condition> ::= <Attribute> = <Attribute>`: True
`<Condition> ::= <Attribute> AVAILABLE <Pattern>`: False
`<Condition> ::= <Condition> AND <Condition>`: True

Frage 20

Richtig

Erreichte Punkte 1,00 von 1,00

Isolation means a transaction executes as if it were the only transaction running.

- ☒ Wahr
☐ Falsch

Die richtige Antwort ist 'Wahr'.

Frage **21**

Richtig

Erreichte Punkte 1,00 von 1,00

Even serializable schedules may cause an inconsistent database state from an application's point of view if no integrity constraint is defined.

- ☒ Wahr ✓
- ☐ Falsch

Die richtige Antwort ist 'Wahr'.

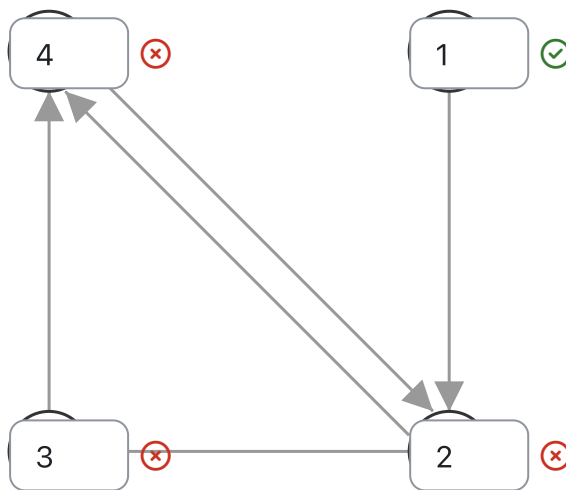
Frage **22**

Teilweise richtig

Erreichte Punkte 0,25 von 1,00

Consider the following schedule: W2(C), W1(A), R3(B), W4(B), R2(C), W3(C), R3(A), W2(B)

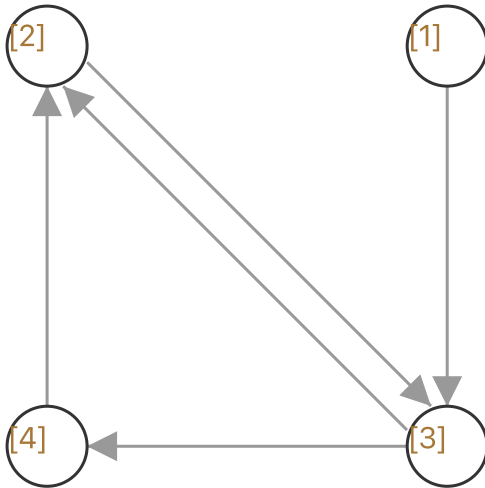
Complete precedence graph by choosing correct entries from dropdowns.



.
Sie haben 1 richtig ausgewählt.

.
Die richtige Antwort lautet:
Consider the following schedule: W2(C), W1(A), R3(B), W4(B), R2(C), W3(C), R3(A), W2(B)

Complete precedence graph by choosing correct entries from dropdowns.

Frage **23**

Richtig

Erreichte Punkte 1,00 von 1,00

Let T1 and T2 be two transactions that are contained in an interleaved schedule.

Let A and B be two different database elements.

Do the following operations conflict or not? Select the appropriate option.

Conflict	Do not conflict			
☒	☐	W1(A) and R2(A)	☒	

W1(A) and R2(A): Conflict

Frage **24**

Richtig

Erreichte Punkte 1,00 von 1,00

The following schedule is conflict-serializable:

R2(B), R4(B), R4(A), W3(C), W2(A), W1(B), W2(B), R3(A)

☐ Wahr☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage **25**

Richtig

Erreichte Punkte 1,00 von 1,00

In practice, **locking-based systems** use strict two-phase locking (2PL) because it leads to maximum parallelization and maximum throughput.

☐ Wahr☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage **26**

Richtig

Erreichte Punkte 1,00 von 1,00

Consider that in addition to the **shared (S)** and **exclusive (X)** locks, your database system has two more locks:

1. **Increment (I)** - increments a database element by a constant, and
2. **Multiplication (M)** - multiplies a database element by a constant.

Fill in the appropriate value for the drop-downs in the compatibility matrix:

(N.A. stands for not applicable. Only the drop-down in the table is graded.)

	Shared (S)	Exclusive (X)	Increment (I)	Multiplication (M)
Shared (S)	Yes	No	N.A.	N.A.
Exclusive (X)	No	No	N.A.	N.A.
Increment (I)	No ✓	N.A.	N.A.	N.A.
Multiplication (M)	No ✓	N.A.	N.A.	N.A.

Frage **27**

Richtig

Erreichte Punkte 1,00 von 1,00

The following schedule is possible under strict two-phase locking (S2PL): R1(D), R2(C), R2(A), R1(A), R3(B), R3(B), R1(B), R3(B)

- ☒ Wahr ✓
- ☐ Falsch

Die richtige Antwort ist 'Wahr'.

Frage 28

Teilweise richtig

Erreichte Punkte 0,75 von 1,00

Consider that you have three transactions, T_1 , T_2 , and T_3 that operate on three database elements A , B , and C . The timestamps of the transactions are as follows: $T_1 = 250$, $T_2 = 275$, $T_3 = 175$, while the initial values of the read and write timestamps of the three database elements is 0.

Consider the following schedule: R1(B), R2(A), R3(C), W1(B), W1(A), W2(C), W3(A)

The goal of the exercise is to maintain a table of operations in the schedule and the action(s) taken once the operation is executed. A few entries in the table have already been filled.

Fill out the missing values in the table below following the algorithm in the Complete Book (Section 18.8.4 The Rules for Timestamp-Based Scheduling):

Operation Action

R1(B)	RT(B)=250
R2(A)	RT(A)=275
R3(C)	RT(C)=175
W1(B)	WT(B)= 250  , C(B)= FALSE 
W1(A)	ROLLBACK
W2(C)	WT(C)= 275  , C(C)= FALSE 
W3(A)	ROLLBACK

Assumptions:

1. The last occurrence of a transaction in the schedule **also includes the commit request** to the scheduler (e.g., for a schedule R3(A), R1(B), W2(C), W1(C) $\rightarrow T_1$ will attempt to commit along with the request R1(B), T_2 will attempt to commit along with the request W2(C), and T_3 will attempt to commit along with the request R3(A)). Familiarize yourself with the action taken by the scheduler once a transaction tries to commit following the algorithm.
2. RT(X) refers to the read timestamp of a database element X
3. WT(X) refers to the write timestamp of a database element X
4. C(X) refers to the commit bit of a database element X

Frage **29**

Richtig

Erreichte Punkte 1,00 von 1,00

Strict Two-phase locking allows more concurrency than Snapshot Isolation (SI).

☐ Wahr☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage **30**

Richtig

Erreichte Punkte 1,00 von 1,00

Match the failure below to the type of failure:

Computer running DBMS crashes - ✓

Die richtige Antwort lautet:

Match the failure below to the type of failure:

Computer running DBMS crashes - [System Failure]

Frage **31**

Richtig

Erreichte Punkte 1,00 von 1,00

No-steal policy means that the database system **allows** a transaction to overwrite the most recent committed value of a database element on disk even though the transaction has not committed.

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage **32**

Richtig

Erreichte Punkte 1,00 von 1,00

Write-Ahead Logging (WAL) ensures atomicity but not durability.

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage **33**

Richtig

Erreichte Punkte 1,00 von 1,00

The analysis phase in recovery starts from the second last checkpoint in the log.

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage **34**

Richtig

Erreichte Punkte 1,00 von 1,00

Consider the following log records. The table contains the log sequence number (LSN), the transaction ID (TID), the previous log sequence number of the transaction (prevLSN), and the action of the log entry. The format of the UPDATE action is: (Page ID, Element ID, Old value, New value).

LSN TID prevLSN ACTION

1	1		BEGIN_TRANSACTION
2	2		BEGIN_TRANSACTION
3	3		BEGIN_TRANSACTION
4	1	1	UPDATE (3, 1, "a", "A")
5	3	3	UPDATE (3, 2, "b", "B")
6	2	2	UPDATE (3, 3, "c", "C")
7			BEGIN_CHECKPOINT
8	1	4	UPDATE (2, 4, "d", "D")
9	3	5	UPDATE (1, 5, "e", "E")
10	1	8	COMMIT
11	1	10	END_TRANSACTION
12			END_CHECKPOINT
13	2	6	UPDATE (3, 6, "f", "F")
14	2	13	UPDATE (2, 7, "g", "G")

The recovery process starts at LSN 7 with the following transaction table and dirty pages table.

Transaction table:

TID lastLSN

1	4
2	6
3	5

Dirty page table:

PID recLSN

3	4
---	---

In the transaction and dirty page tables below, enter the LSN for each transaction/page after the analysis pass. Select "No Entry" if a transaction/page is not in the transaction/dirty page table.

Transaction table:

TID lastLSN

1



2



3



Dirty page table:

PID recLSN

1



2



3



Frage **35**

Richtig

Erreichte Punkte 1,00 von 1,00

The undo pass in recovery goes forward in the log from the last checkpoint.

☐ Wahr

☒ Falsch

Die richtige Antwort ist 'Falsch'.

Frage **36**

Richtig

Erreichte Punkte 1,00 von 1,00

Consider the following log records. The table contains the log sequence number (LSN), the transaction ID (TID), the previous log sequence number of the transaction (prevLSN), and the action of the log entry. The format of the UPDATE action is: (Page ID, Element ID, Old value, New value).

LSN	TID	prevLSN	ACTION	Undo	Pass
1	1		BEGIN_TRANSACTION		
2	2		BEGIN_TRANSACTION		
3	3		BEGIN_TRANSACTION		
4	1	1	UPDATE (1, 1, "a", "A")		
5	2	2	UPDATE (2, 2, "b", "B")		
6	1	4	UPDATE (1, 3, "c", "C")	Undo	✓
7			BEGIN_CHECKPOINT		
8	1	6	UPDATE (1, 4, "d", "D")	Undo	✓
9	2	5	UPDATE (3, 5, "e", "E")	Ignore	✓
10	2	9	COMMIT		
11	2	10	END_TRANSACTION		
12			END_CHECKPOINT		
13	3	3	UPDATE (2, 6, "f", "F")	Undo	✓
14	3	13	UPDATE (3, 7, "g", "G")	Undo	✓

The recovery process starts at LSN 7 with the following transaction table and dirty pages table.

Transaction table:

TID	lastLSN
1	6

TID lastLSN

 2 5

 3 3

Dirty page table:

PID recLSN

 1 4

 2 5

Select the action taken during the undo passes for the UPDATE actions in the drop-down menus in the log above.

Die richtige Antwort lautet:

Consider the following log records. The table contains the log sequence number (LSN), the transaction ID (TID), the previous log sequence number of the transaction (prevLSN), and the action of the log entry. The format of the UPDATE action is: (Page ID, Element ID, Old value, New value).

LSN	TID	prevLSN	ACTION	Undo Pass
1	1		BEGIN_TRANSACTION	
2	2		BEGIN_TRANSACTION	
3	3		BEGIN_TRANSACTION	
4	1	1	UPDATE (1, 1, "a", "A")	
5	2	2	UPDATE (2, 2, "b", "B")	
6	1	4	UPDATE (1, 3, "c", "C")	[Undo]
7			BEGIN_CHECKPOINT	
8	1	6	UPDATE (1, 4, "d", "D")	[Undo]
9	2	5	UPDATE (3, 5, "e", "E")	[Ignore]
10	2	9	COMMIT	
11	2	10	END_TRANSACTION	
12			END_CHECKPOINT	
13	3	3	UPDATE (2, 6, "f", "F")	[Undo]
14	3	13	UPDATE (3, 7, "g", "G")	[Undo]

The recovery process starts at LSN 7 with the following transaction table and dirty pages table.

Transaction table:

TID	lastLSN
1	6
2	5
3	3

Dirty page table:

PID	recLSN
1	4
2	5

Select the action taken during the undo passes for the UPDATE actions in the drop-down menus in the log above.

Frage 37

Richtig

Erreichte Punkte 1,00 von 1,00

The redo pass in recovery goes forward in the log from the maximum flushed log sequence number so far.

☐ Wahr

☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage **38**

Richtig

Erreichte Punkte 1,00 von 1,00

Consider the following log records. The table contains the log sequence number (LSN), the transaction ID (TID), the previous log sequence number of the transaction (prevLSN), and the action of the log entry. The format of the UPDATE action is: (Page ID, Element ID, Old value, New value).

LSN	TID	prevLSN	ACTION	Redo Pass	
1	1		BEGIN_TRANSACTION		
2	2		BEGIN_TRANSACTION		
3	3		BEGIN_TRANSACTION		
4	1	1	UPDATE (1, 1, "a", "A")	Redo	✓
5	2	2	UPDATE (2, 2, "b", "B")	Redo	✓
6	2	5	UPDATE (1, 3, "c", "C")	Redo	✓
7			BEGIN_CHECKPOINT		
8	2	6	UPDATE (1, 4, "d", "D")	Redo	✓
9	2	8	UPDATE (3, 5, "e", "E")	Redo	✓
10	2	9	COMMIT		
11	2	10	END_TRANSACTION		
12			END_CHECKPOINT		
13	3	3	UPDATE (2, 6, "f", "F")		
14	3	13	UPDATE (3, 7, "g", "G")		

The recovery process starts at LSN 7 with the following transaction table and dirty pages table.

Transaction table:

TID	lastLSN
1	4

TID lastLSN

2	6
---	---

3	3
---	---

Dirty page table:

PID recLSN

1	4
---	---

2	5
---	---

Select the action taken during the redo passes for the UPDATE actions in the drop-down menus in the log above. Assume that pageLSN is always less than LSN.

Die richtige Antwort lautet:

Consider the following log records. The table contains the log sequence number (LSN), the transaction ID (TID), the previous log sequence number of the transaction (prevLSN), and the action of the log entry. The format of the UPDATE action is: (Page ID, Element ID, Old value, New value).

LSN	TID	prevLSN	ACTION	Redo Pass
1	1		BEGIN_TRANSACTION	
2	2		BEGIN_TRANSACTION	
3	3		BEGIN_TRANSACTION	
4	1	1	UPDATE (1, 1, "a", "A")	[Redo]
5	2	2	UPDATE (2, 2, "b", "B")	[Redo]
6	2	5	UPDATE (1, 3, "c", "C")	[Redo]
7			BEGIN_CHECKPOINT	
8	2	6	UPDATE (1, 4, "d", "D")	[Redo]
9	2	8	UPDATE (3, 5, "e", "E")	[Redo]
10	2	9	COMMIT	
11	2	10	END_TRANSACTION	
12			END_CHECKPOINT	
13	3	3	UPDATE (2, 6, "f", "F")	
14	3	13	UPDATE (3, 7, "g", "G")	

The recovery process starts at LSN 7 with the following transaction table and dirty pages table.

Transaction table:

TID	lastLSN
1	4
2	6
3	3

Dirty page table:

PID	recLSN
1	4
2	5

Select the action taken during the redo passes for the UPDATE actions in the drop-down menus in the log above. Assume that pageLSN is always less than LSN.

Frage 39

Richtig

Erreichte Punkte 1,00 von 1,00

What is the purpose of checkpoints for logging?

Wählen Sie eine Antwort:

- ☒ Reduce the number of log-records to process during recovery. ✓
- ☐ Guarantee isolation of transactions.
- ☐ Improve the performance of logging.

Your answer is correct.

Die richtige Antwort ist: Reduce the number of log-records to process during recovery.

Frage **40**

Richtig

Erreichte Punkte 1,00 von 1,00

Amdahl's Law implies that beyond a certain number of processors, adding more processors has diminishing returns.

- ☒ Wahr ✓
- ☐ Falsch

Die richtige Antwort ist 'Wahr'.

Frage **41**

Richtig

Erreichte Punkte 1,00 von 1,00

According to Amdahl's Law, in a best-case scenario, the theoretical speedup of a program is equal to the number of processors available.

- ☒ Wahr ✓
- ☐ Falsch

Die richtige Antwort ist 'Wahr'.

Frage **42**

Richtig

Erreichte Punkte 1,00 von 1,00

Which parallel architecture does a multiprocessor laptop use?

Wählen Sie eine Antwort:

- ☐ Shared-disk
- ☐ None
- ☒ Shared-memory ✓
- ☐ Shared-nothing

Your answer is correct.

Die richtige Antwort ist: Shared-memory

Frage **43**

Richtig

Erreichte Punkte 1,00 von 1,00

Besides round robin, hash-based partitioning also guarantees an equal amount of tuples per partition.

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage **44**

Richtig

Erreichte Punkte 1,00 von 1,00

Choosing the wrong partitioning key in horizontal partitioning can lead to data skew and performance issues.

- ☒ Wahr 
- ☐ Falsch

Die richtige Antwort ist 'Wahr'.

Frage **45**

Richtig

Erreichte Punkte 1,00 von 1,00

Vertical partitioning is most useful when the full table is frequently used in queries.

- ☐ Wahr
- ☒ Falsch 

Die richtige Antwort ist 'Falsch'.

Frage **46**

Richtig

Erreichte Punkte 1,00 von 1,00

Which of the following correctly describes a scenario where both replication and fragmentation are used?

Wählen Sie eine Antwort:

- ☐ It is a method where all data is replicated across multiple sites.
- ☐ The entire database is stored at every site, improving performance
- ☒ A relation is partitioned into multiple fragments, and identical replicas of each fragment are stored at different sites. ✓
- ☐ A single copy of the database is split among sites based on availability requirements.

Your answer is correct.

Die richtige Antwort ist: A relation is partitioned into multiple fragments, and identical replicas of each fragment are stored at different sites.

Frage **47**

Richtig

Erreichte Punkte 1,00 von 1,00

Inter-operator parallelism allows multiple operators to run concurrently.

- ☒ Wahr ✓
- ☐ Falsch

Die richtige Antwort ist 'Wahr'.

Frage **48**

Richtig

Erreichte Punkte 1,00 von 1,00

Inter-query parallelism allows multiple queries to execute concurrently.

- ☒ Wahr ✓
- ☐ Falsch

Die richtige Antwort ist 'Wahr'.

Frage **49**

Richtig

Erreichte Punkte 1,00 von 1,00

A transaction coordinator is responsible for maintaining a recovery log at each site in the distributed system.

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.